

Functional Annotation Report: *eutC* (Q88QF2) — Ethanolamine ammonia-lyase small subunit, *Pseudomonas putida* KT2440

Gene/Protein Identity Verification

Field	Value
UniProt	Q88QF2
Gene	<i>eutC</i> (ordered locus PP_0542)
Organism	<i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950)
Protein	Ethanolamine ammonia-lyase (EAL) small subunit; EC 4.3.1.7
Length	272 aa
Domains	Pfam PF05985 (EutC) ; InterPro EutC (IPR009246), EutC_N (IPR042255), EutC_C (IPR042251)
Partner gene	<i>eutB</i> (PP_0543), EAL large subunit (Q88QF1)

Verification passed. The gene symbol *eutC*, the EutC/PF05985 domain set, the EC number 4.3.1.7, and the immediately adjacent *eutB* large-subunit gene all agree with the UniProt-provided identity. The literature on "EutC" as the ethanolamine ammonia-lyase small subunit is directly on-target; no gene-symbol ambiguity was encountered. Direct biochemical/structural characterization exists for the *E. coli* and *Salmonella* orthologues; the *P. putida* protein is annotated by homology (HAMAP rule MF_00601), and I add a *P. putida*-specific genomic-context analysis below.

1. Summary (answer to the research question)

EutC is the small (β) subunit of ethanolamine ammonia-lyase (EAL, EC 4.3.1.7), an adenosylcobalamin (coenzyme B12)-dependent enzyme. Working as an obligate heterodimer with the large subunit EutB, EAL catalyzes the deamination of ethanolamine to **acetaldehyde + ammonia** by radical chemistry. EutC itself does not contain the catalytic TIM barrel; rather it partners with EutB to complete the cobalamin-binding pocket at the subunit interface and is required for a functional enzyme. In *P. putida* KT2440 the enzyme lets the cell use ethanolamine as a carbon and nitrogen source (given exogenous/endogenous B12), operating as a **soluble cytoplasmic enzyme** within a minimal, microcompartment-independent ethanolamine-utilization gene cluster.

2. Primary function: the catalyzed reaction and substrate specificity

Reaction (EC 4.3.1.7, Rhea:15313):



EAL is the committed, first catabolic step of ethanolamine degradation. Comparative genomics states plainly that "ethanolamine-ammonia lyase, the enzyme that breaks ethanolamine into acetaldehyde and ammonia, is encoded by the gene tandem *eutBC*" (Tsoy et al., 2009,).

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The products feed central metabolism: the ammonia serves as a nitrogen source, and acetaldehyde is oxidized (by the neighboring aldehyde dehydrogenase, below) toward acetyl-CoA/acetate as a carbon and energy source.

Substrate specificity. The physiological substrate is ethanolamine (2-aminoethanol). The family is characterized as deaminating "various vicinal amino-alcohols to oxo compounds" (UniProt FUNCTION, MF_00601), i.e. it is a vicinal-amino-alcohol ammonia-lyase; ethanolamine is the canonical and best substrate. The enzyme also deaminates **both enantiomers of 2-amino-1-propanol** (2-aminopropanol), non-stereospecifically; enantiomer-bound crystal structures rationalize the lower affinity for the (R)-form via a Val- α 326 side-chain

conformational change and predict pro-S hydrogen abstraction at C1 (Shibata et al., 2011, [P 21142024](#)),

consistent with ^{15}N -isotope-effect/EPR studies on ethanolamine and (R/S)-2-aminopropanol (Poyner et al., 2006, [P 16734439](#)).

The specificity is thus narrow — small 1,2-amino alcohols. Crystallographic work shows the substrate is held by six hydrogen bonds to active-site residues, with Arg(α 160) contributing binding through the substrate O1 hydroxyl (Shibata et al., 2010, [P 20519496](#))

— the spectator hydroxyl group is mechanistically essential (Wetmore et al., 2002, [P 12440904](#)).

Note on division of labor. The catalytic and substrate/cobalamin-binding residues lie on the **large subunit EutB**. Site-directed mutagenesis shows Arg160 of EutB is critical for both oligomer assembly and radical catalysis (R160A is inactive; R160K lowers k_{cat}/K_M ~180-fold) (Sun et al., 2008, [P 18444665](#)).

This reinforces that **EutC is the non-catalytic structural partner** that completes the interfacial cobalamin site.

3. Mechanism and cofactor (radical, B12-dependent catalysis)

EAL is a **class II (adenosylcobalamin-dependent) isomerase/eliminase**. Its cofactor is **adenosylcob(III)alamin** (coenzyme B12; ChEBI:18408), which UniProt notes "binds between the large and small subunits."

The chemistry is radical-based, not hydrolytic: 1. Homolytic cleavage of the cobalt–carbon bond of AdoCbl generates a **5'-deoxyadenosyl radical** (Frey, 2001, [P 11395404](#)).

2. This radical abstracts a hydrogen atom from ethanolamine, producing a substrate radical. 3.

The substrate radical undergoes a **1,2-amino-group shift** to a carbinolamine radical, which collapses to acetaldehyde + ammonia (Wetmore et al., 2002,).

[P 12440904](#)

Experimental EPR of purified EAL confirms "the formation of substrate-derived radicals during catalysis and the binding of cobalamin in the base-on mode," i.e. with 5,6-dimethylbenzimidazole coordinating the cobalt (Akita et al., 2010,).

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4. Structural role of EutC (the small subunit) and localization within the enzyme

Quaternary structure. The functional unit is a **EutB·EutC heterodimer** ($\alpha\beta$) that assembles into a **heterododecamer (EutB)₆(EutC)₆** — "the enzyme existed as the heterododecamer $\alpha_6\beta_6$ " (Akita et al., 2010,).

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Structurally it is a "trimer of the ($\alpha\beta$)₂ dimer," with "6 large subunits form[ing] a core ring with 6 small subunits projecting outwards" (UniProt SUBUNIT; Shibata et al., 2010).

What EutC does structurally. The catalytic machinery — the (β/α)₈ TIM barrel that holds the active site — resides in the **large subunit EutB**. The **small subunit EutC "covers the lower part of the cobalamin that is bound in the interface of the α - and β -subunits"** (Shibata et al., 2010,).

[P 20519496](#)

Thus EutC: - forms the obligate protein partner of EutB (both genes are required for activity), - helps build and cap the cobalamin-binding site at the subunit interface, - is present in 1:1 stoichiometry with EutB, consistent with the finding that binding one cobalamin-analog inhibitor per $\alpha\beta$ unit completely inhibits the enzyme (Akita et al., 2010).

Domain architecture / evolution. EutC arose by recruitment of a **Rossmann-fold domain fused to an N-terminal α -helical domain** (EutB is the paired TIM-barrel + N-terminal helical subunit) (Tsoy et al., 2009,

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— matching the InterPro EutC_N/EutC_C two-domain layout of Q88QF2. The N-terminal region is dispensable for catalysis: N-terminally truncated EAL retains catalytic properties, and "a short N-terminal sequence is sufficient to change the solubility and stability of the enzyme" (Akita et al., 2010).

5. Subcellular localization

EAL is a **cytoplasmic (soluble) enzyme**. In enterobacteria such as *Salmonella* and *E. coli*, EAL is encapsulated inside a **bacterial microcompartment (metabolosome)** — a proteinaceous organelle whose shell (e.g., EutL,

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confines the volatile/toxic intermediate acetaldehyde (Sturms et al., 2015,

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UniProt's HAMAP-inferred "Bacterial microcompartment" localization for Q88QF2 is transferred from these organisms.

However, a *P. putida* KT2440-specific caveat applies. Genomic-neighborhood analysis (this work) shows the KT2440 *eut* cluster is minimal — *eutC* (PP_0542), *eutB* (PP_0543), an ethanolamine transporter (PP_0544, EutH-like), an aldehyde dehydrogenase (PP_0545, *aldB-I*, EutE-like), and a σ 54-dependent regulator (PP_0546) — and **contains no microcompartment shell-protein genes**. A **genome-wide** UniProt proteome search (this work) confirms this: Pfam **PF00936 = 0 hits**, Pfam **PF03319 = 0 hits**, keyword "carboxysome" = 0, and keyword "microcompartment" returns only *eutC/eutB* themselves. Therefore *P. putida* KT2440 builds **no** bacterial microcompartment, and EutBC functions as a **free cytoplasmic enzyme** rather than inside a BMC. The HAMAP-inferred microcompartment localization is an over-transfer from enterobacteria and should be regarded as inapplicable to this organism.

Cofactor supply is intrinsic. Although UniProt phrases activity as requiring "external vitamin B12," KT2440 encodes cobalamin biosynthesis/salvage genes (*cobN* PP_3507, *cobS* PP_1681, *cobD* PP_1675, *cobE* PP_3409) and — critically — the corrinoid adenosyltransferase **PduO (PP_1349, EC 2.5.1.17)** that generates **adenosylcobalamin**, the exact AdoCbl coenzyme EutBC requires, plus periplasmic cobalamin transport proteins (PP_2418; HutB PP_0524). The organism can therefore both import and adenosylate B12 to feed EAL.

6. Pathway context in *P. putida* KT2440

EutC functions in **amine degradation / ethanolamine catabolism** (UniProt PATHWAY). The reconstructed minimal pathway from the gene cluster:

1. **Uptake:** ethanolamine enters via the EutH-like transporter (PP_0544).
2. **Deamination:** cytoplasmic **EutBC (EutC = PP_0542 small subunit)** converts ethanolamine → acetaldehyde + NH₃ using AdoCbl.
3. **Aldehyde oxidation:** the EutE-like aldehyde dehydrogenase (PP_0545, EC 1.2.1.3) oxidizes acetaldehyde, channeling carbon toward acetyl-CoA/acetate and central metabolism.
4. **Nitrogen assimilation:** released NH₃ is assimilated as a nitrogen source.
5. **Regulation:** a σ_{54} -dependent transcriptional regulator (PP_0546) controls cluster expression, consistent with the σ_{54} (RpoN)/NtrC regulatory cascade shown to govern *eut* loci in other proteobacteria (Barnes et al., 2024,

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and linking induction to nitrogen status.

6. **Cofactor dependency:** activity requires the coenzyme adenosylcobalamin; KT2440 supplies it via cobalamin biosynthesis/salvage plus the PduO corrinoid adenosyltransferase (PP_1349), and can also import B12 (PP_2418, HutB).

Ethanolamine itself is a widespread nutrient derived from turnover of the membrane phospholipid **phosphatidylethanolamine** (Kaval & Garsin, 2018,

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catabolizing it lets *P. putida* scavenge additional carbon and nitrogen from its (soil/rhizosphere and host-associated) environment.

This "stripped-down," microcompartment-free architecture is consistent with the variable *eut* gene content across bacteria, including loss/absence of auxiliary genes in many lineages (Tsoy et al., 2009,

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7. Supported and refuted hypotheses

Supported - EutC is the small subunit of AdoCbl-dependent EAL (EC 4.3.1.7) catalyzing ethanolamine → acetaldehyde + NH₃. (UniProt;

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- EutC is an obligate structural partner of EutB, capping the interfacial cobalamin site in an (EutB)₆(EutC)₆ dodecamer.

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- Catalysis proceeds via AdoCbl Co-C homolysis → 5'-deoxyadenosyl radical → 1,2-amino shift.

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- EutC = Rossmann-fold + N-terminal α-helical domain; N-terminus dispensable for catalysis.

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- Substrate range is narrow (small 1,2-amino alcohols): ethanolamine (physiological) plus both enantiomers of 2-aminopropanol; catalytic residues (Arg160) reside on EutB, confirming EutC's structural (non-catalytic) role.

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Refuted / corrected for this organism - The HAMAP-inferred "bacterial microcompartment" localization is **not** supported for *P. putida* KT2440. A genome-wide proteome scan finds **no** BMC shell-protein genes (Pfam PF00936 = 0, PF03319 = 0; "carboxysome" = 0), so EutC acts in the **cytoplasm**. (genome-wide bioinformatic analysis; contrast with

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- The "requires external vitamin B12" wording is only partially apt: KT2440 encodes cobalamin biosynthesis/salvage plus the PduO adenosyltransferase (PP_1349) that makes the AdoCbl coenzyme, so cofactor provision is intrinsic. (*bioinformatic analysis*)

8. Limitations and future directions

- Direct biochemical/structural data are from *E. coli*/*Salmonella* EAL; **no *P. putida*-specific experimental study of EutC was found** (literature searches returned none). The *P. putida* protein is annotated by strong homology (HAMAP) and reconstructed pathway logic, not by dedicated in-vitro study — assignments should be read as high-confidence inference.
- Confirmation that *P. putida* KT2440 grows on ethanolamine (as C and/or N source) and the B12 dependence would benefit from targeted growth/knockout experiments; the σ^{54} -regulator and inducer are candidates for regulatory study.
- An AlphaFold/experimental structure of the *P. putida* EutBC complex could confirm the interfacial cobalamin-capping role of EutC in this organism.

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